Functional Representation of the Riemann Zeta Function via Triple-Exponential Towers

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Abstract

This research presents a highly aesthetic and precise numerical model for the Riemann Zeta function $\zeta(s)$ using a nested power tower structure. By defining Akinda's Constant (A_s), we establish a direct transcendental link between $\zeta(s)$ and the fundamental constants ϕ, π, and e.

1 The Representation Model

The core of this research is the following power tower representation, which provides a symmetrical and elegant structure for calculating $\zeta(s)$ for $s > 1$:

$$\zeta(s) = \left(\left(\phi^{\frac{1}{\pi}}\right)^{\frac{1}{\phi}}\right)^{\frac{A_s}{e}} \tag{1}$$

Where:

- $\phi = \frac{1+\sqrt{5}}{2} \approx 1.61803398$ (The Golden Ratio)
- $\pi \approx 3.14159265$
- $e \approx 2.71828182$ (Euler's Number)
- A_s is Akinda's Constant, specific to the input s.

2 Special Closed-form for $\zeta(3)$

A remarkable discovery within this model is the specific relation for $s = 3$ (Apéry's Constant). The empirical data suggests a high-precision closed-form for A_3 as follows:

$$A_3 \approx \frac{\pi \cdot e}{\phi} \tag{2}$$

Substituting this into the main representation (Eq. 1) yields a direct transcendental approximation for $\zeta(3)$:

$$\zeta(3) \approx \left(\left(\phi^{\frac{1}{\pi}}\right)^{\frac{1}{\phi}}\right)^{\frac{\pi}{\phi}} \tag{3}$$

3 Comprehensive Data Table

The constants A_s were empirically derived to ensure a numerical precision of 10^{-16} compared to standard Zeta values.

Table 1: Computed Values of Akinda's Constant (A_s)

Input Variable (s)	**Akinda Constant** (A_s)
1.05	86.841896210679720
1.1	67.747717016565837
1.5	27.573030324597450
1.61803 (ϕ)	23.131831111181775
2.0	14.291015100363020
2.5	8.435591835018455
3.0	5.284381743097510
3.5	3.426258316303970
4.0	2.271571397337800
4.5	1.529414851122160
5.0	1.041239337498470
5.5	0.714757443577792
6.0	0.493721928232680
6.5	0.342685968259390
7.0	0.238746958019560
7.5	0.166823879965915
8.0	0.116839710132640
8.5	0.081983759118460
9.0	0.057611424388890
9.5	0.040532693048230
10.0	0.028544169944260

4 Asymptotic Behavior for $s > 10$

For values where $s > 10$, Akinda's Constant follows an exponential decay model, allowing for a simplified calculation of $\zeta(s)$:

$$\zeta(s) \approx \left(\left(\phi^{\frac{1}{\pi}}\right)^{\frac{1}{\phi}}\right)^{\frac{A_{10}\cdot 2^{(10-s)}}{e}} \tag{4}$$

Where $A_{10} = 0.02854416994426$.